

VirtualCDU Masterplan v5.0

Mission: Turn VirtualCDU from a functional browser-based FMC/MCDU trainer into a measured, testable, high-fidelity training tool for Boeing 737 NG procedures first, then Airbus A320 MCDU workflows, with strong visual realism, robust procedural behavior, reliable MSFS integration, and a sustainable development/testing process.

This masterplan combines the refined accuracy plan, the deeper app analysis, repository findings, and an additional research pass into one unified roadmap. The uploaded refined plan remains the strongest baseline: it defines Boeing 737 NG visual accuracy as the highest priority, requires Phase 0 measured baselines before treating scores as facts, and already includes current status estimates, real hardware references, phased work, exit gates, pilot review, iPad performance targets, and MSFS validation goals [151†L1-L19] .

1. North Star

VirtualCDU should become a **procedure-first FMC/CDU trainer** that feels close enough to real hardware to support credible practice, while remaining honest about its limits. It should not pretend to be a certified training device. The target product identity should be:

“A high-fidelity, web-based Boeing 737 NG CDU trainer with Airbus A320 MCDU support, built for realistic preflight, route, performance, navigation, and simulator-connected practice.”

The strongest differentiator is not just “a CDU-looking UI.” It is the combination of:

1. **Realistic page flow**
2. **Measured visual fidelity**
3. **Correct input validation and scratchpad behavior**
4. **Guided tutorial workflows**
5. **SimBrief/navdata realism**
6. **MSFS connected mode**
7. **iPad/PWA cockpit usability**

The public README already positions VirtualCDU as a Boeing 737 NG FMC/CDU simulator that works on desktop and iPad, supports offline/no-install usage, offers guided tutorials, and aims to connect to MSFS 2020 for live CDU sync. It also lists Boeing pages, extra Boeing pages, Airbus MCDU pages, touch-optimized behavior, and input validation as product features.

2. Product principles

Principle 1: Boeing first, Airbus second

The refined plan correctly prioritizes Boeing 737 NG fidelity. The Boeing side already has the strongest conceptual spine: IDENT → POS INIT → RTE → DEP/ARR → PERF INIT → THRUST LIM → TAKEOFF REF, plus LEGS, PROG, HOLD, FIX, CLB, CRZ, DES, DIR INTC and N1 LIMIT. Airbus should remain supported,

but the project should avoid pretending Airbus is equally complete until its FMGS-specific behavior, color semantics, and page logic mature.

Principle 2: Measured realism over guessed realism

Current scores should remain estimates until Phase 0 replaces them with evidence. The refined plan is right to demand baseline screenshots, reference-photo sets, color ΔE measurement, character-position variance, coverage reports, iPad performance profiling, and pilot review rubrics 【151†L1-L19】 .

Principle 3: Every page should be either functional, display-only, or explicitly out of scope

Avoid “fake interactivity.” If a page’s LSK appears selectable, the action should either work, show a correct scratchpad message such as `NOT SUPPORTED` , or be disabled. The current Implementation Status notes known remaining gaps where Airbus pages like PERF APPR, FUEL PRED, SEC F-PLN, RAD NAV, and DATA INDEX are static/display-only in CONTROL mode ¹ , while PMDG round-trip verification and broader E2E coverage remain incomplete.

Principle 4: Training value depends on workflows, not just screens

The app should not only render pages. It should teach why each entry exists, how pilots cross-check it, what can go wrong, and what the CDU/MCDU should do in response.

Principle 5: Connected mode is a product of its own

MSFS integration must be treated as a separate reliability product. SimConnect is designed for external components to communicate with Microsoft Flight Simulator, so integration requires a dedicated connection layer, diagnostics, latency tracking, mapping profiles, and live validation matrix.

3. Source-of-truth hierarchy

Use this hierarchy to resolve conflicts:

1. **This masterplan** – strategic direction and phase structure.
2. **FMS_ACCURACY_IMPROVEMENT_PLAN_REFINED.md** – current authoritative project plan and measured targets 【151†L1-L19】 .
3. **IMPLEMENTATION_STATUS.md** – current implementation tracker, but confirm whether the local file or GitHub version is newest. Some sections note 43 unit tests and 5 E2E tests, while the refined plan reports 54 unit tests and 9 E2E tests ¹ .
4. **Reference hardware photos, Boeing/737 references, Airbus references, pilot review** – truth for visual and procedural fidelity.
5. **Automated test results** – truth for regression confidence.
6. **Live MSFS sessions** – truth for integration validity.

4. Current-state synthesis

What is strong today

Area	Current strength
Core architecture	Monorepo with shared logic, frontend, and backend separation.
Display model	Page renderers produce FMC display data and LSK mappings.
Boeing workflow	Full Boeing preflight path is implemented conceptually.
Validation	ICAO, flight number, altitude, speed, QNH, V-speed, and hold/fix validation are present or planned.
Tutorial engine	Guided scenarios, highlighting, error counts, hints, and performance tracking exist or are planned.
Testing	TypeScript, unit tests, E2E tests, and build verification are part of the workflow.
PWA/iPad intent	Touch targets, offline usage, and iPad cockpit mounting are central to the product.
MSFS direction	PMDG SimConnect scaffolding exists; FBW/Fenix should remain explicitly scoped or out-of-scope.

What is still weak

Area	Main gap
Visual fidelity	Colors, font, spacing, bezels, button depth, key grouping, and screenshot comparison are not yet measured.
Navigation realism	Lightweight parser and limited nav dataset are not enough for real procedure realism.
Airbus depth	Several Airbus pages are display-only/static in CONTROL mode ¹ .
Connected mode	PMDG keypress → CDU update → display readback still needs live validation.
E2E coverage	Full preflight, DEP/ARR selection, LEGS editing, HOLD/FIX, SimBrief import, and CONTROL mode need tests.
Training depth	Tutorials need full-flight scenarios, error correction, SOP roles, and scoring.
Documentation	User manual, developer architecture guide, and scope/license notes are missing.

5. Master roadmap

Phase 0 — Evidence baseline and plan lock

Duration: 3–5 days

Priority: Mandatory before further subjective scoring

Goal: Convert assumptions into measured baselines.

Tasks

1. Status reconciliation

2. Confirm local status: TypeScript, unit tests, E2E tests, build output, audit status.
3. Resolve mismatch between refined plan baseline and GitHub status.
4. Update `IMPLEMENTATION_STATUS.md` with one current truth.

5. Screenshot baseline

6. Capture repeatable screenshots for at least 35 screens, not just 25.
7. Include Boeing pages (IDENT, POS INIT, RTE, DEP, ARR, LEGS, PERF INIT, THRUST LIM, TAKEOFF REF, CLB, CRZ, DES, DIR INTC, N1 LIMIT, HOLD, FIX, PROGRESS, MENU, FAIL/OFF annunciation, scratchpad invalid entry, EXEC lit state) and Airbus pages (INIT A, INIT B, F-PLN, DEP/ARR, PERF TAKEOFF, PERF APPR, PROG A, RAD NAV, SEC F-PLN, DATA INDEX, MCDU MENU).
8. Include integration/UI screens: connection diagnostics (disconnected, connecting, connected), tutorial step active, error/hint state, iPad portrait/landscape.

9. Reference library

10. Collect real 737 NG CDU photos by model/manufacture.
11. Label references by source, aircraft variant, CDU generation, display type, and image quality.
12. Collect Airbus MCDU references separately.

13. Measurement method

14. Define screenshot normalization: crop region, scale, camera angle correction, luminance correction, anti-aliasing tolerance.
15. Define color sampling method: sampling points, display calibration assumptions, ΔE calculation, tolerance.
16. Define character grid measurement: line height, character width, baseline position, left/right alignment.
17. Define acceptable non-realism: browser font smoothing, iPad glare, non-certification disclaimer.

18. Pilot review rubric

19. Create a form with scoring for visual realism, page flow realism, scratchpad behavior, Boeing procedure accuracy, Airbus procedure accuracy, iPad usability, training usefulness, error-message clarity, overall confidence.

20. Testing baseline

21. Run coverage using Vitest; run Playwright E2E baseline tests and create screenshot snapshots for major screens.

Exit gates

- One authoritative status table exists.
- Screenshot capture is reproducible.
- Reference library exists.
- Pilot review rubric exists.
- Coverage baseline exists.
- No unmeasured score is presented as a factual score.

Phase 1 — Boeing 737 NG visual fidelity

Duration: 2–3 weeks

Priority: Highest

Goal: Make the Boeing CDU look and feel recognizably close to real 737 NG hardware.

Workstream 1.1: Display engine

Implement field-level display semantics and a **colour-token system**. Real CDUs use multiple colours to convey meaning: titles are cyan or boxed, active guidance and targets are magenta, tuned VOR/navaid data is green, normal/entered data is white, and pending modifications (when the EXEC key is lit) appear in a shaded or inverse white. Mandatory fields and cautions should use amber, and error or failure messages should use red. Define a unified `DisplayColor` union (for both Boeing and Airbus palettes) and map each semantic role to a calibrated hex value and Tailwind class. Provide helper functions (e.g. `getColorClass`, `getColorHex`, `isValidColor`) so page renderers can attach a colour to each `DisplayLine`. This token system makes it easy to swap palettes for Airbus while preserving semantic meaning.

Workstream 1.2: Font and character grid

- Select or create a **realistic FMC-style monospaced font**. Adjust letter-spacing and line-height so each character sits in a fixed cell.
- Investigate using a **canvas-based display renderer** for pixel-perfect control: a `<canvas>` allows drawing each line with exact spacing, optional scanlines and glow, and can avoid browser font smoothing artefacts.
- Define 24 columns × 14 rows explicitly and ensure left/right labels and line body align exactly.
- If a canvas is used, optimise redraws (only repaint changed lines) to maintain ≥ 55 fps on iPad. Regardless of rendering method, add tests for line overflow, truncation and correct placement.

Workstream 1.3: Physical CDU shell

Improve bezel geometry, straight screen edges, screen inset depth, keycaps, key spacing, key depression, per-key backlighting, brightness knob placeholder or functional brightness control, and

iPad-mounted proportions. Add **subtle screen glow or scanline effects** to mimic the phosphor look of real CDUs. Implement ripple feedback on touch devices so the display feels tactile. Tighten spacing between the display and keys to better match real hardware proportions. Consider a slight bevel and realistic aluminium-like styling for the outer bezel.

Workstream 1.4: Function/navigation keys

Verify layout, label, and behavior for all 14 function/navigation keys (INIT REF, RTE, CLB, CRZ, DES, DIR INTC or MENU, LEGS, DEP ARR, HOLD, PROG, N1 LIMIT, FIX, PREV PAGE, NEXT PAGE). Ensure each key is clearly labelled and that missing keys are added. **Refine button styling:** add subtle gradients, borders and a small backlight glow for active or pressed keys; implement smooth press animations with a slight scale and shadow change; and include a “key-wobble” or depth effect so presses feel tactile. For the LSKs, show context-sensitive labels instead of generic arrows when space allows, so the function of each LSK is clear.

Workstream 1.5: Visual regression

Use Playwright snapshot tests for major screens.

Workstream 1.6: Placeholders and cursor

Implement **context-sensitive placeholders** on data entry fields. Real CDUs use different placeholder formats depending on the expected input: alphabetic fields show `----`, numeric fields show `...` or `—`, decimal fields show `...`, and flight-level fields show `FL---`. Display these placeholders in shaded or inverse white to indicate that data is required. Add a **blinking underscore cursor** on the active input line or scratchpad so pilots can see exactly where text will appear. These small details make typing feel closer to real hardware and help students understand what type of data is expected.

Exit gates

- Character-position variance $\leq 4\%$ vs reference set.
- Color $\Delta E \leq 4$ for major Boeing display colors.
- All 14 function/navigation keys verified visually and functionally.
- ≥ 55 fps on target iPad profile.
- Pilot visual realism $\geq 4.0/5$ with at least four reviewers.
- No TypeScript/test/build regressions.

Phase 2 — Boeing procedural fidelity and page behavior

Duration: 3–4 weeks

Priority: High

Goal: Make Boeing pages behave like an FMC trainer, not a static page emulator.

Workstream 2.1: Preflight flow completion

Make the Boeing flow robust by implementing realistic missing-data prompts, V-speed deletions when dependent values change, route discontinuity behavior, and EXEC staging for route edits.

Workstream 2.2: LEGS page

Implement insertion before/after selected waypoint, deletion, speed/altitude constraint modification, discontinuity handling, and realistic EXEC staging.

Workstream 2.3: HOLD page

Add hold fix entry, inbound course, turn direction, leg time/distance, EFC time placeholder, hold available in route context, EXEC staging, “not enough data” message, and tutorial step.

Workstream 2.4: FIX page

Add reference fix, radial/distance rings, multiple entries, display reference rings on simple route map or textual preview, and constraint cueing for descent planning.

Workstream 2.5: DES, CRZ, CLB and PROG pages

Implement top-of-climb and top-of-descent estimates, basic VNAV path model, DES NOW prompt, DRAG REQUIRED/THRUST REQUIRED approximation, cruise altitude step-climb placeholder, fuel/time/distance predictions based on simplified model.

Workstream 2.6: Approach/landing support

Add or expand the approach reference page: landing flap selection, VREF entry, ILS frequency/course display, runway length/distance placeholder, landing tutorial segment.

Exit gates

- Full Boeing preflight tutorial completes without workaround.
- $\geq 90\%$ of common pilot entry errors produce correct scratchpad responses.
- LEGS, HOLD, FIX, DES, and approach reference are test-covered.
- Pilot procedure realism $\geq 4.2/5$.

Phase 3 — Airbus MCDU polish and scoped functionality

Duration: 2–3 weeks

Priority: Medium-high

Goal: Make Airbus mode coherent, visually correct, and honestly scoped.

Workstream 3.1: Airbus visual semantics

Implement Airbus-specific display semantics: white titles/advisories, blue modifiable fields, green active/non-modifiable data, amber mandatory/warning messages, magenta constraints/managed targets.

Workstream 3.2: Airbus function keys

Verify key labeling and grouping (DIR, PROG, PERF, INIT, DATA, F-PLN, RAD NAV, MCDU MENU, AIRPORT and OVFY if included).

Workstream 3.3: Airbus page support

Define each Airbus page as functional, display-only, or out-of-scope. Aim to make INIT A, INIT B/ FUEL PRED, F-PLN, DEP/ARR, PERF TAKEOFF, PERF APPR, PROG A and RAD NAV functional; make SEC F-PLN and DATA INDEX display-only if necessary.

Workstream 3.4: Airbus CONTROL mode

Resolve static pages either by wiring backend state and handlers or by marking them display-only in the UI. The implementation status notes these pages may currently be static in CONTROL mode ¹.

Exit gates

- Airbus color $\Delta E \leq 5$.
- Airbus page labels/key layout pass pilot review.
- All visible Airbus LSK actions either work or are intentionally disabled.
- Pilot Airbus visual rating $\geq 4.0/5$.

Phase 4 — Navigation database, ARINC-lite, and SimBrief

Duration: 4–6 weeks

Priority: High after Boeing visual/procedural work

Goal: Move from a small mock dataset to realistic route/procedure training.

Workstream 4.1: Nav database architecture

Build an expandable schema for airports, procedures and legs.

Workstream 4.2: Data sources

Evaluate Navigraph's Navigation Data API (which is designed for external applications like mobile apps and updates according to AIRAC cycles) and open-source nav datasets for integration.

Workstream 4.3: ARINC-lite procedure support

Implement simple ARINC 424 constructs: track to fix, direct to fix, course to fix, initial fix, runway fix, hold fix, discontinuity insertion, speed/altitude constraints, procedure transition selection.

Workstream 4.4: SimBrief import

Implement SimBrief route import: origin/destination, alternate, cost index, cruise altitude, fuel and ZFW, SID/STAR matching, mismatch warnings.

Workstream 4.5: Route fixture set

Create at least 20 versioned route fixtures across domestic/international and Boeing/Airbus scenarios for testing and tutorial use.

Exit gates

- $\geq 90\%$ of fixture routes load successfully.
- At least 20 real SimBrief routes covered.

- SID/STAR selection works for supported fixtures.
- LEGS page reflects procedure legs and constraints.
- Nav database schema documented.
- Pilot procedure realism $\geq 4.2/5$.

Phase 5 — MSFS connected mode hardening

Duration: 3–4 weeks

Priority: High

Goal: Make connected mode reliable enough for regular simulator use.

Workstream 5.1: Connection architecture

Implement explicit connection states (disconnected, connecting, connected, degraded, reconnecting, error), heartbeat, timeout, exponential backoff, last message timestamp, message queue, dropped-message counter, latency measurement, adapter capability detection.

Workstream 5.2: PMDG round-trip validation

Create a validation matrix covering different MSFS versions and verifying keypress send, CDU page change, display readback, scratchpad readback, failure states and recovery.

Workstream 5.3: Adapter profiles

Implement an adapter abstraction for different aircraft types with capabilities enumeration (key input, display readback, scratchpad readback, aircraft state, route sync).

Workstream 5.4: FBW/Fenix decision gate

Decide whether Airbus FBW/Fenix integration is out-of-scope or requires a dedicated mapping phase.

Workstream 5.5: Integration test harness

Build a mock SimConnect adapter to emulate key input, display readback, latency and disconnect/reconnect for CI tests.

Exit gates

- PMDG keypress → CDU update → display readback verified in live simulator.
- Connection success $\geq 98\%$ across test matrix.
- Average display latency < 80 ms in connected sessions.
- Mock integration tests run in CI.
- Connected mode has clear diagnostics UI.
- Pilot sim training value $\geq 4.3/5$.

Phase 6 — Training system depth

Duration: Parallel after Phase 2, then ongoing

Priority: High for user value

Goal: Convert VirtualCDU into a structured learning tool.

Workstream 6.1: Tutorial curriculum

Create levels from CDU basics to full-flight scenarios, covering preflight, takeoff/climb, cruise, descent, approach/landing, and abnormal scenarios. Provide cross-checks, SOP roles, and interactive hints.

Workstream 6.2: Scoring

Score correct actions, time to complete, number of invalid entries, number of hints used, cross-check completion, EXEC timing, discontinuity resolution.

Workstream 6.3: Crew coordination

Add PF/PM role labels for tasks like performance entry, route entry and cross-check. Simulate multi-pilot procedures.

Workstream 6.4: Adaptive hints

Provide multiple levels of hints, from subtle cues to step-by-step instructions.

Exit gates

- Full preflight tutorial completion rate $\geq 85\%$ without external help.
- Average confidence gain $\geq +1.5$ points.
- Pilot training effectiveness $\geq 4.3/5$.
- Tutorial tests cover every scenario.

Phase 7 — Testing, CI and quality gates

Duration: Ongoing; first major push 1–2 weeks

Priority: Critical for maintainability

Goal: Make future changes safe.

Unit tests

Add tests for V-speed cross-field validation, HOLD staged edits and EXEC commit, LEGS delete mode, LEGS insert waypoint, constraint editing, FIX radial/distance validation, DES NOW logic, route discontinuity logic, Airbus unsupported actions, scratchpad error clearing, PMDG adapter mock behavior.

E2E tests

Add Playwright tests for full Boeing preflight flow, DEP/ARR selection, LEGS editing, HOLD creation and EXEC, FIX radial/distance entry, SimBrief import mock, tutorial completion, CONTROL mode mock adapter, iPad viewport layout, and visual regression snapshots.

Coverage targets

- Shared FMC logic $\geq 85\%$.
- Store actions $\geq 80\%$.
- Server FMCEngine $\geq 85\%$.
- Page renderers $\geq 90\%$.

- Adapter mocks $\geq 75\%$.

CI gates

Every PR must pass TypeScript checks, unit tests, E2E smoke tests, coverage threshold, visual regression for changed pages, build, lint, audit policy check.

Exit gates

- E2E coverage includes all major flows.
- Coverage thresholds enforced.
- Visual regression integrated into CI.
- Mock MSFS adapter integration tests pass.

Phase 8 — PWA, offline, and iPad cockpit mode

Duration: 1–2 weeks

Priority: Medium-high

Goal: Make VirtualCDU feel like a reliable mounted cockpit device.

Workstream 8.1: Offline behavior

Configure service worker caching for app shell, fonts, display assets, tutorial content, demo navdata; avoid caching live SimConnect data or large navdata updates without versioning.

Workstream 8.2: Kiosk mode

Add fullscreen prompt, wake lock, orientation guidance, iPad safe-area handling, large touch target validation, connection status indicator, offline/online badge, update-available prompt.

Workstream 8.3: Performance

Measure initial load, time to interactive, keypress-to-display latency, animation frame rate, memory usage over 30 minutes, offline startup time.

Exit gates

- App works after refresh offline.
- iPad maintains ≥ 55 fps.
- No visible layout shift during CDU interactions.
- Update flow does not corrupt state.

Phase 9 — Documentation, licensing and project operations

Duration: 1 week initial, then ongoing

Priority: Medium but important

Goal: Make the project maintainable and adoptable.

User documentation

Provide getting-started guides, Boeing preflight tutorial, Boeing in-flight reference, Airbus quickstart, MSFS connection guide, troubleshooting, known limitations, and clear disclaimer that the tool is not for real-world operations.

Developer documentation

Document architecture, shared renderer contract, adding a page, adding an LSK action, frontend vs backend state rules, CONTROL mode flow, SimConnect adapter interface, test strategy, visual regression process, navdata schema.

Licensing

Review fonts, reference images, navdata, SimConnect libraries, third-party packages, and ensure they are compatible with the project's license.

Governance

Add ADRs for major decisions, `SCOPE.md`, `KNOWN_LIMITATIONS.md`, `TEST_MATRIX.md`, `PILOT_REVIEW_RUBRIC.md`, `ROADMAP.md`, `CHANGELOG.md`.

6. Expanded backlog

Must do

- Phase 0 evidence baseline
- Boeing visual fidelity
- Real screenshot comparison
- Full Boeing E2E preflight
- PMDG live round-trip validation
- Navdata schema
- SimBrief fixtures
- HOLD/FIX tests
- Airbus scope clarification

Should do

- Airbus visual pass
- Airbus PERF APPR inputs
- Airbus RAD NAV tuning
- DES NOW trainer logic
- Approach reference page
- PWA offline robustness
- Visual regression CI
- Pilot review sessions

Could do

- CPDLC/ACARS training page
- Voice-guided tutorials

- Route map overlay
- Advanced scoring
- Failure scenarios
- Multi-user instructor mode
- Export training report
- Hardware keyboard mapping
- Real CDU skin variants by manufacturer

Won't do for now

- Certification as a training device
- Full ARINC 424 all-leg support in first navdata phase
- Full FBW/Fenix integration unless scoped
- Exact proprietary Boeing/Airbus fonts if licensing is unclear
- Real-world operational use claims

7. Master success metrics

Area	Metric	Target
Visual	Boeing character-position variance	$\leq 3\%$ end-state
Visual	Boeing color ΔE	≤ 3 end-state
Visual	Airbus color ΔE	≤ 5 near-term, ≤ 4 later
Performance	iPad frame rate	≥ 55 fps
Testing	Shared FMC coverage	$\geq 85\%$
Testing	E2E major flow coverage	100% of defined flows
Procedure	Boeing standard operations match	$\geq 95\%$ by pilot matrix
Navdata	SimBrief fixture route loading	$\geq 90\%$
Integration	PMDG connection success	$\geq 98\%$
Integration	Display latency	< 80 ms average
Training	Pilot usefulness rating	$\geq 4.5/5$
Training	User confidence increase	$\geq +1.5$ points
Quality	TypeScript	0 errors
Quality	Build/tests	100% pass

8. Immediate next actions

This week

1. Reconcile `IMPLEMENTATION_STATUS.md` with the refined plan's 54/9 test baseline.
2. Create `TEST_MATRIX.md`.
3. Create `PILOT_REVIEW_RUBRIC.md`.
4. Add Playwright screenshot capture command for 35 baseline screens.

5. Add route fixture folder with first five SimBrief-like test cases.
6. Add E2E test for full Boeing preflight.
7. Add unit tests for V-speed validation, HOLD EXEC, and LEGS delete.
8. Build reference-photo library with metadata.
9. Decide whether Airbus CONTROL mode static pages are “accepted display-only” or “Phase 3 required functionality.”
10. Decide whether FBW/Fenix integration is out-of-scope for v1 or gets a dedicated mapping phase.
11. **Quick visual wins:** begin using the colour-token system on key pages (IDENT, LEGS, PERF INIT, RTE) to introduce cyan titles, magenta guidance and shaded modifications.
12. Add any missing function/navigation keys and improve button hover/pressed states with gradients, shadows and a small backlight glow.
13. Implement context-sensitive placeholders and a blinking cursor on active entry fields.

Next two weeks

1. Implement calibrated Boeing display tokens.
2. Improve font/character-cell rendering.
3. Add screenshot regression for Boeing pages.
4. Implement HOLD/FIX deeper behavior or explicitly scope trainer-level behavior.
5. Build MSFS mock adapter harness.
6. Add PMDG live test checklist.
7. Expand tutorials beyond preflight into takeoff/climb/descent.

9. Recommended final phase order

1. **Phase 0 — Evidence baseline**
2. **Phase 1 — Boeing visual fidelity**
3. **Phase 2 — Boeing procedural fidelity**
4. **Phase 3 — Airbus scoped polish**
5. **Phase 4 — Navdata + SimBrief realism**
6. **Phase 5 — MSFS connected mode**
7. **Phase 6 — Training curriculum**
8. **Phase 7 — Testing/CI hardening**
9. **Phase 8 — PWA/iPad cockpit mode**
10. **Phase 9 — Documentation/licensing/operations**

The key difference is separating Boeing procedural fidelity from navigation database realism. This prevents the project from jumping into ARINC/navdata complexity before the core trainer behavior is excellent.

10. Final master objective

VirtualCDU v1.0 should provide a visually measured, pilot-reviewed Boeing 737 NG CDU trainer with complete preflight, route, performance, LEGS, HOLD, FIX, climb/cruise/descent, and approach-reference workflows; robust validation; a structured tutorial curriculum; offline iPad usability; and verified PMDG MSFS connected mode. Airbus A320 support should be visually coherent and honestly scoped, with deeper FMGS functionality developed after the Boeing training core reaches high fidelity.

This plan keeps the project ambitious but controlled: first measure, then perfect Boeing visuals, then deepen Boeing behavior, then expand route realism, then harden simulator integration, and only then broaden Airbus and advanced training features.

¹ [raw.githubusercontent.com](https://raw.githubusercontent.com/Reedtrullz/RFMC/main/IMPLEMENTATION_STATUS.md)

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